REVIEW ARTICLE

Zebrafish (*Danio rerio*) in Nepal: natural history, habitats, genetic significance and conservation priorities

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Abstract

The zebrafish (*Danio rerio*) is one of the most widely used vertebrate models in biology, yet the wild fish on which that model rests are poorly documented. The species is native to South Asia, and Nepal lies within its natural range, where it occurs in lowland rivers of the Ganges basin, including the Gandaki (Narayani) and Karnali systems. This review brings together published information on the natural history of wild zebrafish and on their occurrence in Nepal, using fish surveys from the Narayani River in Chitwan, national checklists, studies of captive domestication at Begnas Lake in Pokhara, and population genomic work that sampled wild fish from Nepal, India and Bangladesh. Wild populations show deep genetic structure and carry variation that is missing from laboratory strains. Globally the species is classed as Least Concern, but a national Red Data Book listing records it as Vulnerable in Nepal, and the rivers it inhabits face fishing pressure, siltation and agrochemical pollution. We identify the main gaps in knowledge, which are the lack of systematic distribution records, of habitat measurements and of genetic data from Nepalese sites, and we propose a field and laboratory framework to fill them.

Keywords: *Danio rerio*; zebrafish; Nepal; freshwater biodiversity; Narayani River; Chitwan; wild populations;

1. Introduction

The zebrafish, *Danio rerio*, is a small freshwater cyprinid that was first described from the Ganges River system in the early nineteenth century [1]. It entered the laboratory in the 1970s and 1980s, when the University of Oregon group led by George Streisinger developed it as a genetic model and produced some of the earliest clones of a vertebrate [2]. Its embryos are transparent, develop outside the mother and grow quickly, and a single pair can lay hundreds of eggs, which makes the species practical to study at scale [3]. The publication of the reference

genome in 2013 showed that about 70% of human protein-coding genes have at least one zebrafish counterpart, and that 84% of human genes associated with disease have one [4].

Most of what is known about zebrafish biology comes from a few inbred laboratory strains. The natural history of the species, meaning where it lives, what it eats, when it breeds and how its populations differ, has received far less attention [5, 6]. Nepal lies within the native range of the species [3, 7], and its southern lowland rivers drain into the Ganges, yet published information specific to Nepal is scattered across fish surveys, national checklists and a small number of population studies.

This review has three aims. First, to summarise what is known about the natural history of wild zebrafish and apply it to the Nepalese setting. Second, to collect the records of the species in Nepal and the information on its genetic diversity and conservation status. Third, to identify the gaps that local research could close and to propose a practical framework for doing so. It is a synthesis of published work and contains no new field data.

2. Natural history of wild zebrafish

2.1 Taxonomy and morphology

Danio rerio (Hamilton, 1822) belongs to the minnow and danio group of the family Danionidae [1]. Adults are small. Wild fish are typically 1.8 to 3.7 cm long, and the largest recorded individual, which was kept in captivity, reached 64 mm [6, 7]. The body is pale with several dark blue horizontal stripes separated by gold interstripes that run from behind the head into the caudal fin (Figure 1).

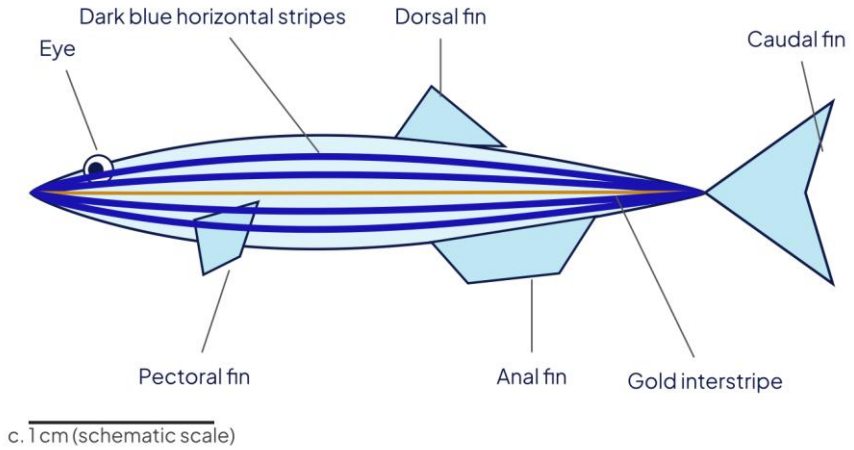


Figure 1 | Schematic of an adult zebrafish, lateral view. Principal external features. The drawing is an illustration for orientation and is not a measured specimen. Typical wild adults are 1.8–3.7 cm long.

2.2 Habitat

Wild zebrafish live mainly in shallow, slow-moving streams and in still pools beside streams. They are reported from rice paddies, drainage ditches and stock ponds, and are especially abundant in silt-bottomed, well-vegetated pools [5, 6]. Reported physical conditions are broad. Water temperatures range from 12 to 39 °C and pH from 5.9 to 9.8, and elevations from about 8 m to 1,576 m [6]. Another source gives records up to 1,795 m [3]. The difference reflects the sources compared and not a measurement made for this review (Figure 3).

2.3 Diet, breeding and lifespan

Zebrafish are omnivorous. They eat insects, small crustaceans, zooplankton, algae, plant material and detritus, and take mosquito larvae readily [6, 7]. Breeding in the wild occurs mainly during the summer monsoon, when temporary pools form, whereas laboratory strains spawn year-round [6]. Spawning tends to take place near daybreak, and males can be territorial. Laboratory fish may live several years, but timing of maturity and lifespan in the wild are not known, and wild fish are generally considered to be short-lived [6, 7].

2.4 Competitors and predators

Other small cyprinids such as *Esomus* and *Puntius*, and larger *Danio* species, are likely competitors. Predators of adults include snakeheads (*Channa*), the knifefish *Notopterus*, catfishes, kingfishers, herons and dragonfly larvae, and many species probably take eggs and larvae [6]. The effects of competition and predation on survival and reproduction in the wild remain unquantified.

3. Zebrafish in Nepal

3.1 Setting

Nepal spans a steep gradient from the Terai lowlands, which lie in the south at roughly 60 to 300 m, through the Churia (Siwalik) range and the middle hills to the high Himalaya (Figure 2b). Three large rivers, the Koshi, the Gandaki (known as the Narayani in the lowlands of Chitwan) and the Karnali, cross this gradient from north to south and join the Ganges in India (Figure 2a). The Terai and the foothills are the part of the country most likely to match the shallow, slow, warm waters in which wild zebrafish are usually found.

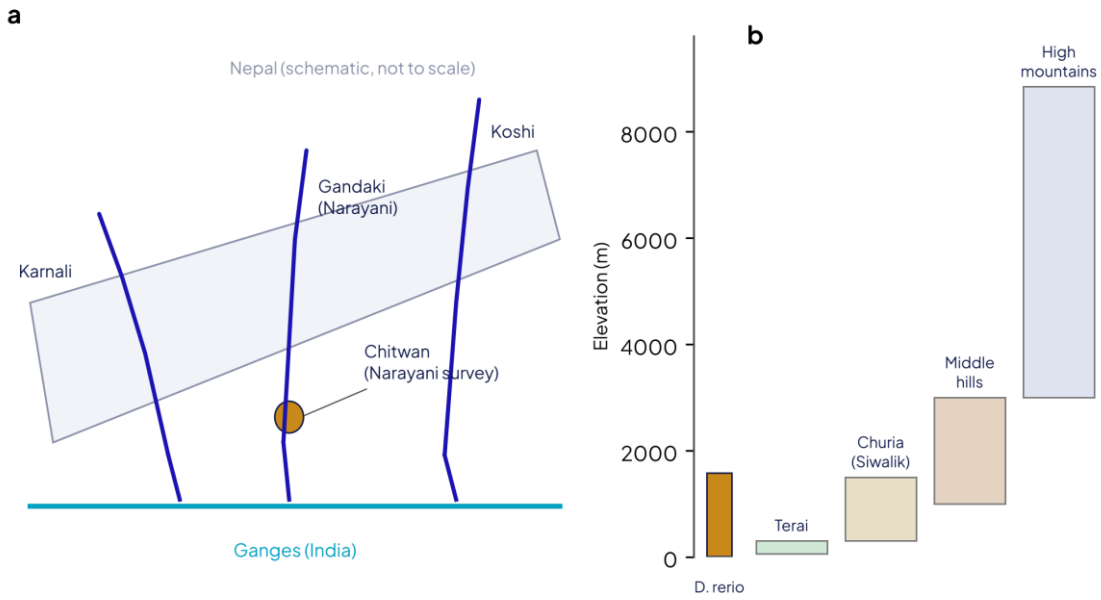


Figure 2 | Drainage and elevation context for zebrafish in Nepal. (a) Schematic drainage of the three major river systems towards the Ganges, with the Narayani survey area at Chitwan marked. The drawing is conceptual and not to scale. (b) Approximate elevation ranges of Nepal's physiographic belts compared with the elevation range reported for wild *D. rerio* (about 8–1,576 m) [ref. 6]. Belt limits are approximate and overlap.

3.2 Records of the species

Records of *D. rerio* in Nepal are few and come from different kinds of source (Table 1). The national checklist compiled in FishBase lists the species from the Gandaki and Karnali rivers, with the local name *sano dedhawa* [8]. A fish survey of the Narayani River system in Chitwan, carried out by the Fisheries Development Centre in Hetauda, reported both *D. rerio*, known locally as *khida*, and *D. devario*, known as *chaiyapothi*, as common in the area. Both were found mostly during the low-water phase, from October to January [9]. The Narayani flows about 332 km through Nepal, its main feeder being the Kaligandaki, and the survey sampled it together with the Trishuli, Rapti and several smaller streams.

A third source comes from Begnas Lake near Pokhara. In a study of native ornamental fishes, *D. rerio* was one of five species identified as suitable for domestication, the others being *Puntius conchonius*, *P. sophore*, *Barilius barna* and *Danio devario*. Fish were collected from an irrigation canal and the outlets of the lake between February and September 2020. The authors noted that *D. rerio* was not caught in some seasons, so its survival in captivity could not be fully assessed [10].

Table 1 | Published records of D. rerio in Nepal.

Source	Location	What was reported
FishBase national checklist [8]	Gandaki and Karnali rivers	Species listed; local name sano dedhawa; Red Data Book status given as Vulnerable
Fisheries Development Centre survey [9]	Narayani River system, Chitwan (about 244 m)	Common; local name khida; mostly caught October–January; found with D. devario, Puntius, Rasbora daniconius and Esomus danricus
Chand *et al.* [10]	Irrigation canal and lake outlets, Begnas Lake, Pokhara	One of five native ornamental species proposed for domestication; not caught in some seasons

The environmental ranges reported for the species across its distribution (Figure 3) give a first idea of what to expect at Nepalese sites, but no standardised measurements of temperature, pH, flow, substrate or vegetation have been published for Nepalese zebrafish habitats in the sources reviewed here.

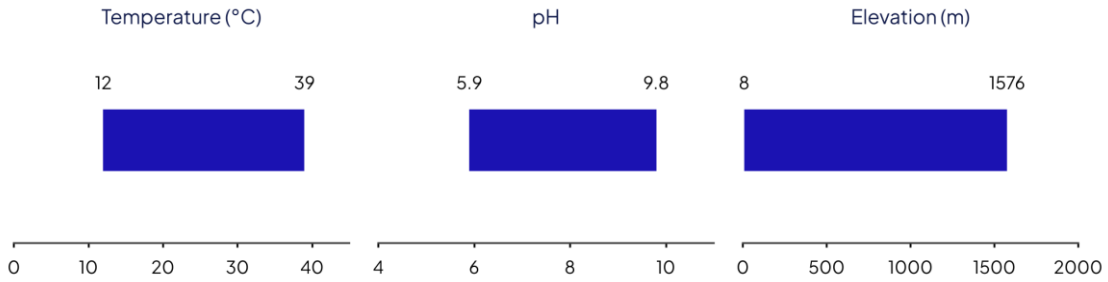


Figure 3 | Environmental ranges reported for wild *D. rerio*. Temperature, pH and elevation ranges compiled for the species across its range, after Parichy [ref. 6]. These are range-wide figures and not measurements from Nepal.

3.3 Associated fishes and the wider fauna

In the Narayani system the fish community around *Danio* includes small, abundant species such as *Puntius*, *Rasbora daniconius* and *Esomus danricus*, which have little commercial value but fill an important place in the food web [9]. *Danio devario* typically prefers faster-flowing water than the zebrafish, which favours the margins of streams and rivers [10]. This habitat difference is one reason why the two species can coexist and why surveys need to record flow and microhabitat as well as presence.

4. The Nepalese fish fauna in context

FishBase lists 202 freshwater fish species in 38 families for Nepal [8]. Other studies cited in the Nepalese literature give about 230 native species [10]. Species richness varies by river, with the Koshi holding 108 species, the Narayani 102, the Karnali 74 and the Mahakali 69 (Figure 4) [8]. More than 130 species occur in snow-fed rivers and mountain lakes. The Narayani survey alone collected 69 species from 9 orders and 19 families, of which 34 were cyprinids, and noted that earlier surveys of the same system had reported between 23 and 111 species depending on the elevation range and method [9]. This spread shows how strongly numbers depend on survey effort.

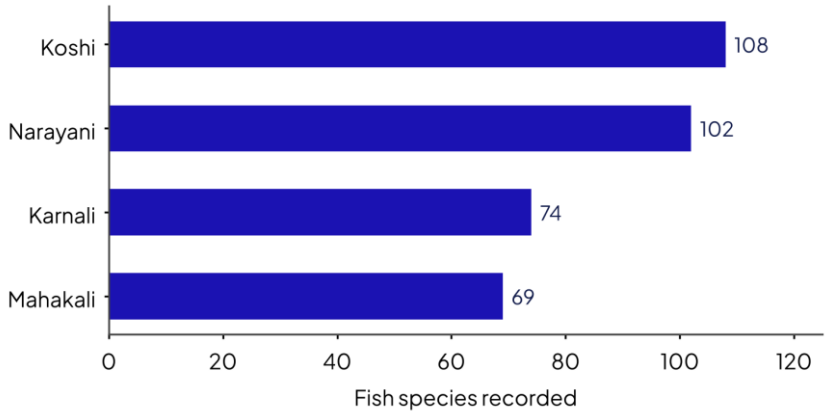


Figure 4 | Fish species richness in four major river systems of Nepal. Counts after Shrestha (1990) as reported in FishBase [ref. 8].

Ten Nepalese species are listed in the Red Data Book, five as Vulnerable and five as Endangered, and most belong to the genera *Labeo*, *Puntius* and *Schizothorax* [8]. *Danio rerio* appears among the species with a Red Data Book entry in the FishBase checklist. The basis and date of that listing could not be verified from the sources consulted here and should be checked against the original national assessment.

5. Genetic diversity and the value of Nepalese populations

The strongest evidence that wild populations matter for science comes from population genomics. Whiteley and colleagues sampled 13 wild populations from Nepal, India and Bangladesh and three commonly used laboratory strains, and typed 1,832 polymorphic single-nucleotide markers and a mitochondrial gene [11]. Wild fish fell into three major mitochondrial clades that differ by about 5.8% on average, and into five genetically distinct groups at nuclear markers (overall FST = 0.170) that corresponded to separate geographic regions. The laboratory strains formed two further groups that were distinct from every wild group. The authors concluded that substantial variation present in the species as a whole is missing from the laboratory strains they analysed.

Other work reinforces the point. Two laboratory genomes differed at about 7 million single-nucleotide positions, and a single wild fish carried more than 5 million variants. After correcting for genome size, zebrafish single-nucleotide variants are roughly four times as frequent as in humans [4, 6]. Wild fish also carry a major sex-determining region on chromosome 4 that has been lost from laboratory strains, which may explain conflicting findings about sex determination in the laboratory [6]. Laboratory strains also differ from one another, so results from one strain may not generalise to the species [6].

These results are relevant to Nepal in two ways. Nepalese populations were part of the sample in the 2011 study, so they contribute to the diversity it describes. And because the geographic groups it identified are distinct, populations from other parts of the range, including Nepalese sites that were not sampled, may carry variation that has not been described. At present there is no published description of the genetic make-up of the zebrafish of the Narayani, Koshi or Karnali basins in the sources reviewed here.

6. Conservation status and threats

6.1 Status

Globally, *D. rerio* is assessed as Least Concern by the IUCN [3]. This reflects its wide range, tolerance of varied conditions and persistence in modified habitats such as paddies and ditches. A species can be secure at global scale and still be locally depleted, and the Red Data Book entry noted above for Nepal indicates concern at the national level [8]. The conservation argument for the zebrafish is therefore less about preventing extinction and more about keeping the habitats and the genetic variation of wild populations intact.

6.2 Freshwater biodiversity in decline

The Living Planet Index tracks 34,836 population trends across 5,495 species. Between 1970 and 2020 monitored wildlife populations fell on average by 73%, and monitored freshwater populations of fish, birds, mammals, amphibians and reptiles fell by 85%, the steepest decline of any environment (Figure 5) [12]. The main threats listed are habitat loss and degradation, overexploitation, invasive species, pollution, climate change and disease. Rivers receive the combined effects of all land uses upstream, so they are exposed to each of these at once.

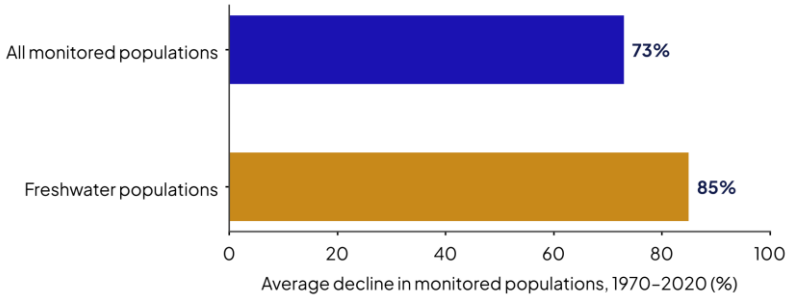


Figure 5 | Average decline in monitored vertebrate populations, 1970–2020. Data from the Living Planet Index 2024 [ref. 12].

6.3 Local pressures

The Narayani survey identified a set of pressures acting on fish in the lowlands of Chitwan (Table 2, Figure 6) [9]. These include unregulated fishing, the use of dynamite and poison, the taking of broodstock during spawning, nets with a mesh smaller than 2 cm and the collection of fry. They also include deforestation leading to erosion and silting, and pollution from industrial waste and agricultural chemicals such as insecticides and pesticides. Seasonal changes in water level and winter drying of small streams add natural stress. Because zebrafish live in the same shallow, warm margins as farmland and villages, they are likely to meet these pressures early.

Table 2 | Pressures on fish of the Narayani River system reported by the Fisheries Development Centre, Hetauda.

Category	Reported pressures
Fishing practices	Unregulated fishing; dynamiting; poisoning; taking of broodstock during spawning; nets of mesh smaller than 2 cm; collection of fry
Land use	Deforestation causing erosion and silting
Pollution	Industrial waste; insecticides; pesticides; piscicides
Natural variation	Seasonal changes in water level; winter desiccation of small streams
Fishing effort	Catch per person per gear per day of 0.9 kg at Gaindakot and 1.3 kg at Bharatpur; 47 active fishing families recorded

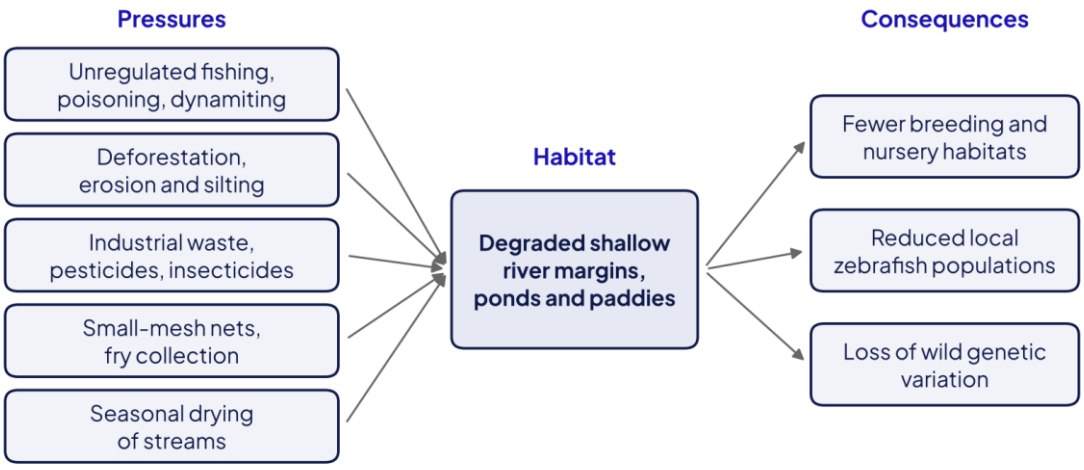


Figure 6 | Conceptual pathway from local pressures to loss of zebrafish habitat and variation. Pressures are those reported in the Narayani survey [ref. 9]. The pathway is an interpretive model and has not been tested with data from Nepal.

7. Gaps in knowledge

The sources reviewed leave several gaps that are specific to Nepal.

- **Distribution.** There is no systematic map of confirmed zebrafish sites by river basin, elevation and season. Existing records are incidental to broader fish surveys.
- **Habitat.** No standardised habitat measurements, such as temperature, pH, flow, substrate and vegetation, have been published for Nepalese zebrafish sites.
- **Genetics.** The genetic composition of populations in the Narayani, Koshi and Karnali basins has not been described, and their relation to the groups found in India and Bangladesh is unknown.
- **Life history.** Timing of breeding, age at maturity and lifespan in Nepalese waters have not been recorded.
- **Status.** The basis of the national Red Data Book entry is unclear, and trends in abundance are not documented.
- **Threat effects.** The effects of agrochemicals, siltation and fishing on zebrafish and related small fishes have not been measured.
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8. A framework for research in Nepal

A coordinated programme can close these gaps at modest cost, because zebrafish are small, easy to sample with fine-mesh nets and found in accessible shallow waters. We propose five linked steps (Figure 7). Sites would be chosen to cover river basins, the elevation gradient and the seasons, with repeat visits during and after the monsoon because wild breeding follows it

[6]. At each site, standard water-quality and habitat variables would be recorded, together with the co-occurring fish community. Fin clips or whole-fish samples would be taken under permit for genetic analysis and compared with published data from India and Bangladesh [11]. Results would be shared with universities, local government and the communities who use the same waters.



Figure 7 | Proposed research workflow for zebrafish in Nepal. The workflow links field survey, habitat measurement, genetic sampling and reporting.

Table 3 | Variables proposed for standard recording at each survey site.

Domain	Variables
Location	Coordinates; river and sub-basin; elevation; date and season
Water	Temperature; pH; dissolved oxygen; conductivity; turbidity; flow and depth
Habitat	Substrate; vegetation cover; connection to paddies, ditches or ponds; land use nearby
Fish community	Presence and abundance of Danio species and associated Puntius, Esomus and Rasbora; predators
Biological samples	Standard length and body condition; fin clips or whole fish for DNA under permit
Pressures	Fishing methods seen; evidence of poisoning or dynamiting; agrochemical use; siltation

Collaboration will strengthen this work. NZINC has research affiliations with Agriculture and Forestry University, Mid-West University and Far-West University in Nepal, and with Duke University and the University of Oregon, where laboratory zebrafish work began. It has signed a memorandum of understanding with Duke University on research into native zebrafish, aquatic biodiversity and river ecosystem conservation, and an agreement with Agriculture and

Forestry University on research and academic exchange (information from the NZINC website, November 2026). Local fishers and farmers are also practical partners, since they know where small fishes occur through the seasons.

Information on the species has uses beyond conservation. Zebrafish from varied natural habitats could widen the range of variation available to biomedical researchers, who rely on a few laboratory strains [6, 11]. The species is also used in drug discovery, where embryo assays can show within about a day whether a compound is selective, against the months required by conventional routes [13]. Documenting and protecting the wild source of this model organism is therefore of scientific interest to Nepal and the wider research community.

9. Conclusions

The zebrafish is native to Nepal's lowland rivers, but it has been studied there far less than in the laboratory. The available evidence shows that the species is present and common in at least parts of the Narayani system, that it is listed in national records from the Gandaki and Karnali rivers, and that its habitats are under pressure. Population genetic work shows that wild zebrafish hold much of the species' variation, which laboratory strains lack. Systematic surveys of distribution, habitat and genetics across Nepal would give a basis for conservation planning, strengthen the link between local research and international science and show what the wild populations of this model organism can still contribute.

Declarations

Data availability. All figures were prepared for this article from published information; Figures 1, 2a, 6 and 7 are conceptual schematics.

Competing interests. The authors are affiliated with NZINC, the publisher of this journal.

References

1. Hamilton, F. *An Account of the Fishes Found in the River Ganges and its Branches* (Archibald Constable, Edinburgh, 1822).
2. Streisinger, G., Walker, C., Dower, N., Knauber, D. & Singer, F. Production of clones of homozygous diploid zebra fish (*Brachydanio rerio*). *Nature* **291**, 293–296 (1981).
3. Zebrafish. *Wikipedia*. https://en.wikipedia.org/wiki/Zebrafish (accessed November 2026). Used as a secondary summary of the primary literature on conservation status, range and research milestones.

4. Howe, K. *et al.* The zebrafish reference genome sequence and its relationship to the human genome. *Nature* **496**, 498–503 (2013).
5. Engeszer, R. E., Patterson, L. B., Rao, A. A. & Parichy, D. M. Zebrafish in the wild: a review of natural history and new notes from the field. *Zebrafish* **4**, 21–40 (2007).
6. Parichy, D. M. Advancing biology through a deeper understanding of zebrafish ecology and evolution. *eLife* **4**, e05635 (2015).
7. Animal Diversity Web, University of Michigan Museum of Zoology. *Danio rerio* (zebra danio). https://animaldiversity.org/accounts/Danio_rerio (accessed November 2026).
8. FishBase. Biodiversity of Nepal: country checklist of fishes, including Red Data Book status and river-wise species richness after Shrestha (1990). https://fishbase.mnhn.fr/country/CountryInfo_Biodiversity.php?c_code=524 (accessed November 2026).
9. Dhital, R. R. Fish survey of the Narayani River system, Chitwan, Nepal. Fisheries Development Centre, Hetauda. Food and Agriculture Organization of the United Nations. https://www.fao.org/4/y3994e/y3994e0e.htm (accessed November 2026).
10. Chand, S., Prasad, A., Regmi, R. S. & Husen, M. A. Identification and domestication of native ornamental fishes of Begnas Lake, Pokhara, Nepal. *Nepalese J. Zool.* **6** (Suppl. 1), 7–11 (2022).
11. Whiteley, A. R. *et al.* Population genomics of wild and laboratory zebrafish (*Danio rerio*). *Mol. Ecol.* **20**, 4259–4276 (2011).
12. Zoological Society of London & WWF. *Living Planet Index and Living Planet Report 2024* (2024). https://www.zsl.org/news-and-events/news/living-planet-index-2024
13. Vanderbilt University Medical Center. Zebrafish swim into drug development. *VUMC Reporter* (2010). https://news.vumc.org/reporter-archive/zebrafish-swim-into-drug-development/